

DESi ablation — is the gain selection, governance, or just structured context?

Falsification-oriented extension of the ab_evidence A/B harness. Real run (claude-sonnet-4.5, temp 0, 3 reps/case); the sample is small, so results are **directional, not a verdict**. No DESi win is claimed beyond what the numbers support.

Backend status: **REAL**. Model-dependent accuracy & degeneration are measured.

Conditions

- A — baseline: the full chat history.
- B — normal DESi: the case's own categorised state slice (id+what per category).
- C — wrong-slice: B's format/budget, but another case's state (cross-domain donor).
- D — status-stripped: B's claim texts with governance metadata removed.
- E — budget-matched status-stripped: D's texts, no metadata, padded with inert filler to B's token budget (~1%).

Input-side characterisation (deterministic, no model)

slice_info_recall = fraction of the case's true ground-truth items actually inside the injected slice. It shows the wrong slice (C) really starves the model of correct information (**0.0**) while B and D carry it (**1.0**); A has the full chat. This describes the **inputs**, not model behaviour.

Core cases

case	condition	input_tokens	slice_info_recall
case1_architecture	A	653	-
case1_architecture	B	736	1.0
case1_architecture	C	613	0.0
case1_architecture	D	444	1.0
case1_architecture	E	743	1.0
case2_research	A	748	-
case2_research	B	655	1.0
case2_research	C	736	0.0
case2_research	D	433	1.0
case2_research	E	662	1.0
case3_debugging	A	697	-
case3_debugging	B	614	1.0
case3_debugging	C	656	0.0
case3_debugging	D	406	1.0
case3_debugging	E	621	1.0
case4	A	4263	-
case4	B	1753	1.0
case4	C	631	0.0
case4	D	1125	1.0
case4	E	1760	1.0

Density sweep (padding grows A; slice constant)

case	condition	input_tokens	slice_info_recall
case6	A	8579	-

case	condition	input_tokens	slice_info_recall
case6	B	3287	1.0
case6	C	631	0.0
case6	D	2057	1.0
case6	E	3294	1.0
case7a_30k	A	26797	-
case7a_30k	B	3287	1.0
case7a_30k	C	673	0.0
case7a_30k	D	2057	1.0
case7a_30k	E	3294	1.0
case7b_60k	A	53196	-
case7b_60k	B	3287	1.0
case7b_60k	C	673	0.0
case7b_60k	D	2057	1.0
case7b_60k	E	3294	1.0

Note: D is ~35–40% smaller than B (that gap IS the governance metadata's footprint). **E** controls for it — D's texts padded with inert filler to B's budget (within ~1%) — so only **B>E** (not merely B>D) would be evidence for the metadata. Runner: temp 0, fixed seed, 3 reps/case.

Model results — recall (mean of 3 reps @ temp 0)

Core cases

case	A	B	C	D	E
case1_architecture	0.833	1.0	0.0	1.0	1.0
case2_research	0.538	0.692	0.077	0.718	0.769
case3_debugging	0.695	0.75	0.0	0.917	0.889
case4	0.857	0.968	0.0	0.976	0.905

Density sweep (case6 unpadded → 30k → 60k padding on A)

case	A	B	C	D	E
case6	0.796	0.718	0.024	0.49	0.616
case7a_30k	0.428	0.753	0.024	0.471	0.612
case7b_60k	0.427	0.702	0.024	0.494	0.612

Degeneration rates — core

condition	loop	bad_frame	coh_no_cont	contra	invalid_reuse
A	0.0	0.0	0.0	0.5	0.0
B	0.08	0.0	0.0	0.0	0.0
C	0.0	1.0	1.0	0.0	8.5
D	0.0	0.0	0.0	0.0	0.0
E	0.0	0.0	0.0	0.0	0.0

Degeneration rates — density

condition	loop	bad_frame	coh_no_cont	contra	invalid_reuse
A	0.0	0.0	0.0	1.22	0.0
B	0.0	0.0	0.0	0.0	0.0
C	0.0	1.0	1.0	0.0	10.67

condition	loop	bad_frame	coh_no_cont	contra	invalid_reuse
D	0.78	0.0	0.0	0.0	0.0
E	0.0	0.0	0.0	0.0	0.0

Findings from THIS run (conservative; n=4 core + 3 density, 3 reps, temp 0, claude-sonnet-4.5)

1. Selection IS load-bearing (strong). Wrong-slice C collapses: mean recall 0.019 (core) / 0.024 (density) vs B 0.853 / 0.724; C shows invalid_claim_reuse $\approx$ 8–11 and bad_framing/coherence-without-continuity = 1.0 (it fluently parrots the wrong case). A plausible-but-wrong slice is far worse than no slice.

2. B-vs-A is mostly long-context robustness. Core B–A $\approx$ +0.122; but in the density sweep B–A runs -0.08 (case6, no padding) $\rightarrow$ 0.33 (30k) $\rightarrow$ 0.27 (60k): at low density A can match/beat B, and B's edge appears and grows only as padding dilutes A. Mainly robustness under context pressure, not a general reasoning gain.

3. Metadata governance: weak, density-dependent evidence. Budget-matched B–E $\approx$ -0.038 on core (B $\approx$ E — metadata NOT shown to help on short cases) but +0.111 across the density cases (B > E consistently, token-controlled). The metadata-free flat list D also loops badly on the long states (loop rate $\approx$ 0.78) while B and E do not. So there is SOME token-controlled evidence the typing helps under pressure, but it is small ($\sim$ 0.1), inconsistent across case types, and not significant at this n.

No empirical DESi win is claimed. Supported: correct selection is essential; DESi's advantage over full chat is largely long-context robustness. Not supported as a general effect: epistemic-metadata governance (only weak, density-dependent signal). Sample is tiny and the evaluator is paraphrase-blind — directional only.

Degeneration metrics (first-class, deterministic, unit-tested)

- **loop_trap** — the answer repeats near-identical units (stuck in a loop)
- **contradiction_persistence** — a known open conflict restated as if settled
- **invalid_claim_reuse** — the answer restates a claim from a known-invalid pool
- **bad_framing_nonrecovery** — share of the answer traceable to a wrong injected slice
- **coherence_without_continuity** — fluent output that preserves little of the true state

What would weaken vs strengthen the DESi claim

Weaken: C $\approx$ B on accuracy (gain is generic structured context); D $\approx$ B (metadata decorative); A $\geq$ B except at extreme density (long-context artefact).

Strengthen: C collapses / high bad_framing_nonrecovery (selection is load-bearing); **B > E** (budget-matched) on conflict/decision typing and degeneration (metadata governs, not token count); B's edge and lower degeneration grow with density (robustness under pressure).

Limitations (stated plainly)

Real backend run (sonnet-4.5, temp 0, 3 reps). Tiny sample (4 core + 3 density) — no significance test; differences of a few items are noise. D token-budget confound now controlled by E. Paraphrase-blind evaluator (relative comparison only). Wrong-slice budget only approximate. Negative results are primary; no promotional claims.

Reproduce: `python ab_evidence/ablation_run.py` (deterministic) · set OPENROUTER_API_KEY then re-run for the full table · `pytest tests/ab_ablation -q`.